

Hamza HOUIR ALAMI

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Portfolio | GitHub | LinkedIn

Research interests: Inventory and supply chain analytics; industrial reliability; forecasting and operational decisions; simulation and optimization under uncertainty.

Education

Undergraduate GPA: **3.89/4.0**

Arts et Métiers Institute of Technology

Lille / Paris, France

Integrated B.S./M.S. in Industrial and Mechanical Engineering

Sept 2025 – Expected Jun 2028

- Top-10 French engineering school; highly selective public Grande École specializing in mechanics, manufacturing, and industrial systems.
- Selected for the US-oriented academic pathway. Relevant studies: mathematics, computer science, machine learning, optimization, production planning, and manufacturing systems.

Lycée Faidherbe

Lille, France

Intensive Preparatory Program for Competitive Admission to Elite Engineering Schools

Sept 2023 – Jul 2025

- Completed a two-year, full-time curriculum in advanced mathematics, physics, mechanics, and engineering science; joined the honors track in Physics and Engineering Sciences, ranked **No. 9 nationally**.

Groupe Scolaire La Résidence

Casablanca, Morocco

French Baccalaureate – Mathematics, Physics & Chemistry, Advanced Mathematics

2023

Work Experience

AMJE Lille – Arts et Métiers Junior-Entreprise

Lille, France

Head of Business Development & Partnerships

Dec 2025 – Present

- Lead **15 business managers**; coordinate client qualification, proposals, commercial follow-up, staffing needs, and partnership development for a student-run consulting firm.
- Generated **EUR 27,000 through 4 signed studies in three months** and expanded the prospecting database from approximately **600 to 1,600 contacts**.

Mohammed VI Polytechnic University – IT Department

Rabat, Morocco

AI & Operations Research Intern

Jul – Aug 2026

- Built a Python/FastAPI decision-support demonstrator connecting data validation, demand forecasting, inventory simulation, and policy recommendations for IT equipment.
- Benchmarked **9 forecasting models** and **6 inventory policies** across **60 department–equipment series**; evaluated **6,480 out-of-sample forecasts** using rolling-origin validation.
- Compared cost and service tradeoffs; on synthetic data, a minimum-stock policy reduced simulated cost by **21.2%** and emergency units by **92.5%** versus corrective purchasing; outlined a multi-period MILP extension.

Safran Nacelles

Casablanca, Morocco

Operations & Manufacturing Engineering Intern

Jun – Jul 2026

- Analyzed tooling flows and assembly-method interfaces; contributed to tooling reorganization, specifications, and CAD work in a nacelle manufacturing environment.
- Studied **36 modeled assembly operations** using representative, anonymized data; combined multicriteria scoring, convex quadratic programming, and constrained numerical optimization.
- Evaluated time-reduction allocations with **5,000 Monte Carlo scenarios**; investigated radiant heating as an alternative to a curing-related waiting cycle under time, energy, and adhesive-quality constraints.

Decathlon

Remote

Product Data Transformation Consultant

Jul – Sept 2026

- Structured migration rules for more than **600 product attributes**, clarifying taxonomy, traceability, consistency requirements, and quality-control checkpoints.
- Defined **4 connected Google Sheets views for 4 user profiles**, including duplicate controls and role-specific navigation.

Academic and Research Projects

Flywheel Shape Optimization and Braking Dynamics

2024 – 2026

Independent extension of preparatory-class research project

Code and reports

- Derived a fixed-perimeter quadratic line-moment bound using Wirtinger's inequality and verified the circular optimum with deterministic Fourier optimization.
- Identified a **28% difference in initial translational kinetic energy** between recorded trajectories; separated passive-inertia and constrained regenerative-braking models in a reproducible workflow.

Compositional French Speech Recognition

2026

Two-person academic project followed by an independent extension

- Built a CNN pipeline using Mel-spectrograms; extended recognition to **0–99** through **22 reusable acoustic components** and grammar-constrained decoding.
- Compared file-based and unseen-speaker evaluation protocols to examine generalization, recording-condition sensitivity, and the limits of headline accuracy.

Glenoid Prosthesis Mechanics and Loosening

2026 – Present

Supervised academic project; parametric modeling and finite-element preparation

- Developing a five-part CATIA/3DEXPERIENCE model of the humeral head, UHMWPE glenoid, cement mantle, and cortical/cancellous bone for Abaqus comparison of geometry, contact, and fixation variants.

Glider Wing – Aerodynamic and Structural Study

2025 – 2026

Supervised two-person engineering project

- Connected analytical wing sizing, Star-CCM+ meshing and CFD, Abaqus structural analysis, and Python post-processing across **3 successive project deliverables**.

Formula Student – FSAM, Arts et Métiers Lille

2025 – 2026

Bill of Materials and Inventory Coordination

- Structured a vehicle bill of materials across **18+ assemblies** in Excel and 3DEXPERIENCE; reconciled required parts with workshop inventory, suppliers, and technical-subteam needs.

Leadership & Volunteering

Student Association of Arts et Métiers Lille

Lille, France

Corporate Relations Officer

Apr 2026 – Present

- Organize employer workshops, networking events, afterworks, and industrial-site visits; maintain direct relationships with Hermès, Safran, L'Oréal, EDF, Alstom, Capgemini, and KPMG.

Atelier des Sciences – STEM Outreach Project

Paris, France

Eight-member science-education team

2026 – Present

- Design reproducible hands-on experiments and filmed demonstrations that use familiar stories to make complex physics accessible to children.

Private Mathematics and Physics Tutor

2024 – Present

Tutored **10 students**, adapting explanations and exercises to individual difficulties and examination goals.

Mentorship for Moroccan Students

2026 – Present

Mentor **5 students** on French scientific pathways, engineering schools, and the transition to studying in France.

Competitive Football

13 years

Played national youth tournaments, served as team captain, and competed internationally at the IberCup Cascais 2018 in Portugal.

Awards & Achievements

Bourse Genin 2026 – Laureate. Excellence and integration scholarship for international students in France.

FC Barcelona Escola Diploma. Completed the academy's youth football training program.

Football Titles. Won a Casablanca regional tournament and the Lycée Faidherbe interclass tournament.

Skills & Interests

Programming and data: Python, SQL, pandas, NumPy, SciPy, scikit-learn, PostgreSQL, Excel.

Analytical methods: Constrained optimization, forecasting, inventory simulation, Monte Carlo methods, CNNs, model evaluation, production planning.

Engineering tools: CATIA / 3DEXPERIENCE, Abaqus, Star-CCM+; CAD, finite-element analysis, CFD.

Languages: Arabic (native), French (fluent), English (advanced).

Interests: Formula 1, geopolitics, football.